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Chapter 11

Biosecurity: Current and Future Strategies

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11.1 Introduction

Aquaculture is the fastest growing food production sector, and its contributions to food, nutritional security and poverty alleviation have been highlighted in Salin *et al.* (Chapter 1, this volume, 2023). It is also a highly diverse production sector. More than 500 species of fish are farmed in an array of environments (freshwater, brackish water, marine), production systems (e.g., ponds, cages, recirculating systems, integrated multitrophic systems, polyculture, nurseries, grow-out facilities, and hatcheries), using three main management strategies (extensive, semi-intensive and intensive) under several different sizes of operation (from backyard and subsistence level to small-, medium- and large-scale (and industrial) operations). Cage aquaculture (see Chapter 1, this volume) contributes significantly to global aquaculture production of several important freshwater and marine finfish species, bridging the supply and demand gap for aquatic foods in the coming decades.

Various diseases and disorders negatively impacting production of cultured finfish, especially in cage culture systems are in this volume. In many instances the lack of efficient and effective biosecurity is the key factor underlying episodes of disease and subsequent production losses. During the last three decades, the aquaculture sector has experienced several re-emerging and newly emerging diseases at a rate of approximately one new disease every three to five years. While measures to prevent and reduce impacts of diseases that affect aquaculture have been launched by many stakeholders (e.g., government, producer, academic sectors), these efforts have not adequately addressed the root causes of disease problems challenging the industry (FAO, 2019).

The aquaculture value chain is complex with many segments and actors operating, with each segment posing a source of risk. As a result, pathogen emergence, re-emergence and spread have, over the past few decades, been a continuous constraint to sustainable aquaculture worldwide.

11.2 Impacts of Diseases in Aquaculture

Disease is the most important threat and hurdle to sustainable aquaculture production. Impacts and losses due to disease are divided into the “predictable”

and the “unpredictable”. While there are costs in treating and managing both, for predictable disease events there are also additional costs associated with prophylactic treatments and management (Shinn *et al.*, 2015). Although major aquatic pathogens and diseases are listed by OIE (Table 11.1; OIE, 2021a), there are many other non-listed pathogens which cause serious losses to aquaculture industry.

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Table 11.1. Key details regarding the OIE-notifiable diseases of fish. For each pathogen, comments on its transmission, persistence in the environment and if a vaccine is available are provided.

Disease	Pathogen	Transmission		Temperature range (°C)	Survival	Vaccine available?	Reference
		Horizontal	Vertical				

Epizootic haematopoietic necrosis	Virus	Yes	No data	11–20	Presumed that the virus can survive months to years on a fish farm and in sediment, plants and farm equipment. There are no data regarding the presence of the virus on eggs and larvae	No	Langdon (1989); OIE (2019a)
Epizootic ulcerative syndrome	Oomycete	Yes	No	Mostly 18–22	Encysted zoospores survive for at least 19 days. Fish eggs and larvae should be disinfected but there is no report of <i>Aphanomyces invadans</i> presence on them	Immunization of snakehead with a crude extract of <i>A. invadans</i> elicited a humoral immune response	Thompson <i>et al.</i> (1997); Lilley <i>et al.</i> (2001); OIE (2019b)
<i>Gyrodactylus salaris</i>	Monogenean	Yes	No	0–25	Detached flukes can survive for 132 h at 3°C to 24 h at 19°C. Eggs should be disinfected	No	Olstad <i>et al.</i> (2006); OIE (2019c)
Infectious salmon anaemia	Virus	Yes	No strong data	–	Over 14 days at 4°C to >10 days at 15°C. Eggs and embryo could be transmission risk	Yes, since 1999	Falk <i>et al.</i> (1997); OIE (2019d)
Infectious haematopoietic necrosis	Virus	Yes	Insufficient evidence	3–18	Virus decay rate is 4.11–4.37 days (8–12°C) in natural seawater and can be inactivated by suspended sediments and the microbial community within the water source. Survival is longer in fresh water than seawater. Transmission on inadequately disinfected eggs possible	Plasmid DNA vaccines containing the glycoprotein gene of the IHNV administered intramuscularly are efficacious	Lapatra (1998); Garver <i>et al.</i> (2005, 2013); Alonso and Leong (2013); Kell <i>et al.</i> (2014); OIE (2021b)
Koi herpesvirus disease	Virus	Yes	Egg-associated transmission cannot be ruled out	opt. 23–25 (>13–<29)	Studies in Israel found the virus survived >4 h and <21 h in water at 23–25°C. Studies in Japan found viral titres dropped within 3 days in environmental water and sediment at 15°C but remained infective for >7 days in filtered water. Virus DNA has been found in river water at 9–11°C, 4 months before an outbreak	A live attenuated virus has been used. Oral administration of a liposome-based vaccine containing inactivated virus has also proven effective	Perelberg <i>et al.</i> (2003); Haramoto <i>et al.</i> (2007); Ilouze <i>et al.</i> (2011); OIE (2019e)
Red sea bream iridoviral disease	Virus	Yes	Not been investigated	25+	Details regarding survival outside the host are unknown	An effective formalin-killed commercial vaccine is able for use in <i>Epinephelus coioides</i> , <i>Epinephelus malabaricus</i> , <i>Pagrus major</i> , <i>Pseudocaranx dentex</i> and <i>Seriola</i> spp.	OIE (2019f)

Salmonid alphavirus infection	Virus	Yes	No convincing evidence/negligible	opt. 12–15	Salmonid alphavirus (SAV) subtype 3 (SAV3) shed by fish was completely inactivated after 3 weeks at 1°C and 4 weeks at 12°C. Graham <i>et al.</i> (2007) indicated the half-life of SAV is 2.5 days at 20°C increasing up to 61 days at 4°C (seawater, no organic loading), 2.5 days (20°C) to 7.7 days (4°C) in seawater with organic loading. In fresh water, SAV half-lives were 2.2 days (20°C) to 7.6 days (4°C) with organic loading, and 5.0 days (20°C) to 13.2 days (4°C) with no organic loading	DNA-based and virus-inactivated commercial vaccines available	Graham <i>et al.</i> (2007); OIE (2019g); Jarungsriapisit <i>et al.</i> (2020)
Spring viraemia of carp	Virus	Yes	Egg-associated transmission cannot be ruled out	>10–<22	5 weeks in river water at 10°C, >6 weeks in pond mud at 4°C, 4 days in pond mud at 10°C	A DNA vaccine that consists of plasmid DNA expression vectors given intramuscularly results in gene expression of immunogenic proteins for KHV in fish muscle tissue	Ahne (1976); Zhou <i>et al.</i> (2014); OIE (2019h)
Viral haemorrhagic septicaemia	Virus	Yes	No	<28	Virus more stable in fresh water (28–35 days at 4°C; 1 year in filtered water at 4°C; 13 days at 15°C) than seawater (10–12 days at 4°C; 4 days at 15°C).	No commercial vaccine. Candidate vaccines, however, have included killed vaccines, attenuated live vaccines, recombinant vaccine in prokaryotic and eukaryotic expression systems, and DNA-based vaccines	Parry and Dixon (1997); Lorenzen and Lapatra (2005); Hawley and Garver (2008); OIE (2021c)

Disease events often pass undetected, undiagnosed, unreported, and/or misdiagnosed thereby hiding the true severity of the disease. If the actual cost pathogens pose to the industry can be estimated, then the information can raise awareness with Governments, and to prioritise research and development programmes focusing on those pathogens. Loss estimates can further help investment in the aquaculture industry when used to calculate the risks for potential investors but also help farmers in their biosecurity and financial contingency plans.

At a farm level, the first critical step is in recognising health issues. Unfortunately, many losses are just accepted as part of normal production. This fatalistic acceptance means that many diseases go unreported, hiding their true severity and threatening the advance of the aquaculture sector.

The impacts of complacency can also be seen in routine farm practices. While the execution of many procedures may be simple, it is human nature to look for short cuts. This happens frequently in biosecurity. Measures are implemented but are then either not maintained or are omitted in the absence of disease episodes and consequentially the probability of a disease with high mortalities increases.

Limited knowledge or understanding about proper biosecurity planning combined with a strong desire for high crop production can, and has, resulted in catastrophic losses (see Figure 11.1). For example, poor site design, high cage density with limited spacing between cages for water exchange and, overstocking of stock can result in poor flush rates, a rapid deterioration in water quality, a rise in nitrogenous wastes, and can facilitate disease transmission. Overstocking results in additional stress on animals kept under culture, reducing their general health condition and predisposition to infections.

[Figure 11.1 – about here]

11.2.1 Economic impacts

Loss estimates for the top 100 diseases impacting on aquaculture production have been compiled (Shinn *et al.*, 2018b) and include a wide range of viral, bacterial, and parasitic pathogens (see Figures 11.2-11.3). Leading viral candidates, include infectious haematopoietic necrosis virus (IHNV) (Saksida, 2006; Garver *et al.*, 2013), infectious pancreatic necrosis virus (IPNV) (Christie, 1997; Munro and Midtlyng, 2011; Jensen and Kristoffersen, 2015), infectious salmon anaemia (ISA) (Rimstad *et al.*, 2011; Kibenge *et al.*, 2013), pancreas disease by salmonid alpha virus (Aunsmo *et al.*, 2010; Pettersen *et al.*, 2015), tilapia lake virus (TiLV; Ferguson *et al.*, 2014; Dong *et al.*, 2017a,b; Surachetpong *et al.*, 2017) and viral haemorrhagic septicaemia (VHS) (Hill, 1992; Dale *et al.*, 2009; Kim *et al.*, 2009). For diseases like TiLV, an orthomyxo-like RNA virus that causes a syncytial hepatitis with a loss of condition, altered swimming behaviour, and high rates of mortality, there is at the present time a general lack of data on losses, as production can readily compensate for losses.

Key bacterial pathogens include *Aeromonas hydrophila* (see Rasmussen-Ivey *et al.*, 2016), *Flavobacterium columnare* (see Shoemaker *et al.*, 2007, 2011),

Piscirickettsia salmonis (see Berger, 2014) and *Vibrio anguillarum* (see Kitao *et al.*, 1983). Likewise, losses resulting from infections of *Streptococcus agalactiae* and *S. iniae* are substantial (Shoemaker *et al.*, 2001; Klesius *et al.*, 2008; Shinn *et al.*, 2018b). Shinn *et al.* (2018b) estimates from reports of *Streptococcus* spp. across Asia, that a 7.5% mortality across the industry can be assumed, except for countries like Bangladesh and the Philippines which harvest fish at smaller size where a 3.75% rate of mortality is applied. Using a figure of 6,192,963 tonnes of 13 farmed species of tilapia (i.e., species and hybrids belonging to the genera *Coptodon*, *Oreochromis*, *Sarotherodon* and *Tilapia*) produced in 2019 (FishStatJ, 2021), then losses due to *Streptococcus* spp. are close to 440,860 tonnes and a harvest value of USD 876.87 million by applying an average value of USD 1,989 per tonne.

Argulus appear to be a leading cause of financial loss due to parasites. In the cyprinid industry, *Argulus*-associated losses are estimated at US\$ 6.4 billion p.a. (i.e., 19.6% of the value of the industry; Shinn *et al.*, 2018b). Other key parasite-related losses include those due to amoebic gill disease (AGD; Munday *et al.*, 2001; Adams and Nowak, 2001; Vass, 2013; Shinn *et al.*, 2015) and, also to salmon lice (i.e., *Lepeophtheirus salmonis* and *Caligus* spp.) which remains the number one parasite problem in salmonid aquaculture with losses of > US\$ 600 million p.a. (Shinn *et al.*, 2015). There are also notable losses due to the monogenetic skin flukes *Benedenia* and *Neobenedenia* which, notably, infect seriolids and can account for 22% of production costs, i.e., approx. US\$ 400 million p.a. (Whittington *et al.*, 2001; Ernst *et al.*, 2002). In fresh water, huge losses of small fish during hatchery-nursery production are due to the ciliate protozoans *Ichthyophthirius multifiliis* and *Trichodina* spp., which together account for an estimated annual loss of USD 56.4 million p.a. assuming a 1% mortality (Shinn *et al.*, 2018b; Shinn *et al.*, Chapter 6, this volume, 2023).

Estimating annual disease losses is challenging, can be variable and dependent on many biotic, abiotic, and anthropogenic factors, and so some cautions should be exercised in the interpretation and application of these figures. By attempting to provide estimates of loss though, it is hoped that this will raise awareness within the industry, help lower the threshold of acceptable loss and drive further improvements in animal welfare, codes of practice, system management and animal health. With a greater transparency on losses associated with disease outbreaks the aquaculture sector could reduce instabilities, improve production, profitability and achieve greater sustainability.

11.3 Minimising Disease Impacts in Aquaculture

Aquaculture expansion and globalisation of aquatic animal trade have caused unprecedented spread of diseases and pathogens. Several aquatic disease epidemics have substantially impacted regional, national, and in some cases international industries and trade (Lafferty *et al.*, 2015). One recent example is the emergence and rapid spread of “early mortality syndrome” or acute hepatopancreatic necrosis disease (AHPND) caused by pathogenic *Vibrio parahaemolyticus* carrying a *Pir*-toxin bearing plasmid in shrimp (De Schryver *et al.*, 2014; Shinn *et al.*, 2018a). This has severely impacted global shrimp production. Other examples include infectious salmon anaemia (ISA) outbreaks in

salmon farms in Norway, Canada, the United States, and Chile (Lafferty *et al.*, 2015) and the spread of epizootic ulcerative syndrome (EUS) in freshwater fish in almost all parts of the world have caused heavy social and economic impacts worldwide (Kamilya and Baruah, 2014).

[Figure 11.2 – about here]

[Figure 11.3 – about here]

Risks to aquaculture, including cage culture, from climate change has been discussed widely (e.g., De Silva and Soto, 2009; Howes *et al.*, 2015; FAO, 2016). Temperature rises above and below the optimal range of tolerance can increase stress on fish due to sub-optimal physiological conditions. Changes in water temperature may result in increased virulence of otherwise dormant pathogens, rampant growth of parasites, and shifts in, and/or extensions to, their distribution. It is also evident that the appearance of harmful algal blooms increases with increasing water temperatures.

Implementation of well-designed and practical biosecurity programmes, both at national and farm-level, are therefore essential to reducing disease outbreaks in aquaculture. Climate change issues, challenges and opportunities must be carefully assessed, analysed, and considered during designing biosecurity strategies and programmes. Such biosecurity programmes, when they are efficiently implemented, the chances of pathogen introduction, establishment and spread could be reduced. A consistent approach for the prevention, control, and eradication of infectious and contagious disease in aquaculture operations, through stringent biosecurity, is therefore necessary, to improve aquaculture sustainability in the coming decades (Palić *et al.*, 2015).

11.4 Biosecurity Principles and Strategies

Biosecurity in aquaculture consists of practices that minimise the risk of pathogen transfer, establishment and spread. These include practices for reducing the stress to fish, thus making them less susceptible to pathogens to disease.

11.4.1 Prevention is better than cure

The implementation of biosecurity measures can help farms and production systems in limiting the entry of undesirable organisms. Barrett *et al.* (2020), in a comprehensive assessment of sea lice infections in the Atlantic salmon, *Salmo salar*, industry concluded that the effective use of barrier technologies such as skirts, snorkels or closed containment systems, coupled with supplementary preventative methods could make delousing treatments unnecessary at many sites, while high-risk locations might require additional management and regulation. They emphasised that preventative methods would be preferable to reactive delousing and moving towards a prevention-focused paradigm on Atlantic salmon farms might improve fish welfare and productivity, while avoiding significant environmental impacts.

Prevention can be achieved, not only by using barriers to pathogens, but also employing other technologies such as genetics and vaccination as part of a comprehensive programme of integrated pathogen management. Enhancing host resistance to infectious disease has received increasing attention as a major goal of farm animal breeding programmes. Selective breeding programmes improving genetic resistance to a range of fish pathogens has already resulted in moderate to high heritability for resistance to viral nervous necrosis (VNN) in European sea bass, *Dicentrarchus labrax* (see Palaiokostas *et al.*, 2018; Griot *et al.*, 2021 a,b) and for resistance to pasteurellosis caused by *Photobacterium damsela* subsp. *piscicida* in gilthead sea bream, *Sparus aurata* (see Palaiokostas *et al.*, 2016; Aslam *et al.*, 2018). Barría *et al.* (2021) discussing the opportunities for identifying TiLV resistance in Nile tilapia, *Oreochromis niloticus* Genetically Improved Farmed Tilapia (GIFT), reported that genetic markers from a quantitative trait locus (QTL) region have potential in marker-assisted selection to improve host resistance, providing a genetic solution to an infectious disease where few other control or mitigation options currently exist. Similarly, research into developing sea louse-resistant salmon has begun; heritable variation exists, and cumulative improvements are reducing susceptibility to sea lice in some salmon lineages (Barrett *et al.*, 2020).

Several effective vaccines against key bacterial and viral pathogens of fish have already been developed and applied in aquaculture (Bøggwald and Dalmo, 2019; Adams and Subasinghe, 2021; see Table 11.1). Successful development of effective vaccines for other pathogens would also reduce the risk of diseases in aquaculture industry.

11.4.2 On-farm versus off-farm biosecurity

Designing and implementing a good biosecurity plan at the farm-level (on-farm biosecurity) is an important step towards protecting farm stocks from disease. New Zealand's Ministry for Primary Industries highlighted seven important steps to be included in daily farm operation process which will help in protecting farm stocks (NZMPI, 2016; Figure 11.4). They are:

- Maintain stock health and welfare
- Be aware of the pathways of risk organisms into, within, and off your farm
- Obtain pest and disease-free stocks
- Keep things clean
- Check your farm
- Report anything unusual
- Have a biosecurity management plan in place

The following are considered important and essential for an effective on-farm biosecurity plan:

- Site design should minimise cross contamination between different sections; permit the management / treatment of water flow through the site; permit each production unit to be effectively isolated and disinfected on a routine basis (see Figure 11.5).
- Use of approved clean sources of seed and feed.

- Establish quarantine procedures for the receipt of new livestock.
- Request health certificates and conduct own health evaluations.
- Written standard operating procedures relating to biosecurity for each hatchery and farm procedure must be in place. There should, for example, also be protocols for routine cleaning and disinfection. All procedures should be communicated to all staff, with training and proficiency monitored where necessary, regularly updated and accompanied by a data log.
- Conduct risk assessments of all procedures.
- Identify control points where procedures could be implemented to prevent, reduce or to remove the risks of disease.
- Biosecurity should be an overall management system. The entire production pipeline must be considered, including a) broodstock / juvenile source/quality; b) stocking densities; c) feed and feeding regimes; d) hatchery disinfection and management; e) water system preparation; f) monitoring of water and soil parameters; g) disease surveillance; and h) training and record keeping.
- Have emergency plans and corrective action plans pre-prepared in advance of a disease outbreak. Each plan should detail the specific actions required for pathogen containment, control, eradication.
- Learn to recognise the signs of disease and / or deviations from normal.
- Conduct health surveillance on a regular basis to monitor the health of hatchery and farm stock and to control on site sanitary status.
- Understand the economic benefits to 'closing the gate' on disease as opposed to the costs of corrective action.
- On discovering a biosecurity breach or disease outbreak, avoid procrastination and act. Avoid reckless action but act appropriately with veterinary support where required.
- Avoid the promiscuous, unsupervised use of medicants to control aquatic pathogens – instead seek proper veterinary advice.
- Conduct regular reviews of biosecurity on-site practices. Revise where necessary. Monitor. Avoid lapses in biosecurity rigor.

While on-farm biosecurity plans help in protecting farm stocks, it should be complemented by an off-farm biosecurity plan, which in actual terms should be a "national biosecurity plan". Such biosecurity plans covering political boundaries help reduce pathogen transfer risks and maintain safe transboundary movement of live animals and their products, provided that the plan is efficiently implemented. Since off-farm biosecurity plans address measures at the border level and beyond, they are bound by relevant international standards and treaties.

There are many examples of comprehensive biosecurity action plans, e.g., Aquaculture Farm Biosecurity Plan: Generic guidelines and template (see awe.gov.au). Risk assessment is one such component. Risk analysis is an internationally accepted standard method for assessing whether trade in a particular commodity, such as a live aquatic animal or its product, poses a significant risk to human, animal, or plant health and, if so, what measures could be adopted to reduce that risk to an acceptable level. Several international factors have spurred the development of risk analysis. They include the liberalisation of international trade through the General Agreement on Tariffs and Trade and the establishment of the World Trade Organization (WTO) and its Agreement on the

Application of Sanitary and Phytosanitary Measures (SPS Agreement). WTO member countries are now required to use the risk analysis process to justify any restrictions on international trade beyond those specified by the Aquatic Animal Health Code (OIE, 2021a). These must be based on risks to human, animal, or plant health.

[Figure 11.4 – about here]

The general framework for an import risk assessment (IRA) for live aquatic animals and their products is laid out in the OIE's Aquatic Animal Health Code (OIE, 2021a). The members of both the OIE and the WTO, are obligated to follow OIE and WTO procedures. Although not obligatory, the International Council for the Exploration of the Sea (ICES) Code of Practice on the Introductions and Transfers of Marine Organisms (ICES, 2005, 2012) also has wide global acceptance and is considered the key framework for assessing proposals to introduce exotic species into new environments outside their native range or to transfer new strains of established species to countries where they are either native or previously introduced. Among other risk analysis frameworks, the ICES Code addresses the evaluation of potential genetic, ecology and pathogen risks associated with the transfer of aquatic organisms.

11.4.3 Emergency preparedness and response

On many occasions, unsuccessful preventative measures have led to the incursion of pathogens into production systems causing disease emergencies. Emergency preparedness has, therefore, been developed and implemented by some, based on experience from unprepared, or incompletely planned, response to disease emergencies. In many instances, this approach has now proven to be economically sustainable, however, learning from these experiences is important. Pre-emergency and contingency investment in insurance, especially for high value species, has become a welcome trend (Subasinghe *et al.*, subm). Disease management success defaulted to pre-disease emergency 'status quo', rather than investment in preparedness to ensure the next outbreak response would be more effective AQ8. Transparent reporting is of utmost importance. However, there is no longer a plausible excuse, with so many international examples as well as guidelines, audit system and shared experiences, to avoid investment in prevention strategies and infrastructure, which can offset the much greater cost of repeated response/emergency approaches (Subasinghe *et al.*, subm.).

[Figure 11.5 – about here]

11.4.4 Disease and pathogen surveillance

Disease surveillance is a critical component of a good biosecurity plan. While some progress exists with aquatic disease surveillance, many barriers remain to the development of surveillance systems that support effective national and farm biosecurity. Although the awareness has been increased and efforts have been made, aquatic animal health services are inadequate in many geographies to design and apply/implement efficient surveillance strategies at national and or regional levels.

Active surveillance is a systematic tool aimed at gathering data to demonstrate the presence or absence of diseases of national concern, to determine their distribution where present, and to detect emerging or exotic diseases that require investigation. Surveillance activities are often used to provide information for disease control programmes, or support health certification requirements, import risk analysis and international trade. Delayed detection of disease outbreaks due to weak surveillance results in higher response costs to control outbreaks or, typically, disease establishment and rapid spread. Improving surveillance and biosecurity requires strengthened government services and regulation. A comprehensive account on the status and prospects of surveillance programmes in reducing the risks of disease in aquaculture is provided elsewhere (Subasinghe *et al.*, *subm.*).

There are number of logistical, practical challenges and considerations to the collection and processing of samples from the field. In addition to the sampling methodologies that may be required in acquiring the target species/specimens and in the physical attributes of the site and access to it, there may be additional environmental, social, ethical, and governance-based issues to address. While many of the following comments have relevance to activities conducted on land, much of the following discussion focuses on the challenges of sampling finfish reared in a diverse range of systems and environments.

For a health surveillance programme to be effective, there is a need to sample and conduct regular checks, but each sampling point should be comprehensive to provide a clear picture of the health status of the fish stock at that moment in time. At the same time, some caution should be exercised in taking accurate single point samples and in the consequential decisions that are made. Depending on the sensitivity and specificity of the diagnostic tests used, or in the suspicion of an early-stage infection where the probability of pathogen prevalence is expected to be low, a larger sample size may be required. This is particularly important in health certification prior to the movements of stock or before the stocking of new systems to ensure stocks and their culture systems are free of infection. A robust programme of on-farm health surveillance is also centered in good sample sizes as a means of the early detection of pathogen threats but also allowing for the implementation of appropriate interventions in a timely manner, as opposed to responding to a crisis where the management options may become limited.

11.4.5 Safe movement of live aquatic animals

Application of good biosecurity at all levels will reduce the risk of pathogen transfer and establishment. As aquaculture has become a global industry requiring the transboundary movements of live animals and animal products, risk analysis for aquatic animal pathogens has become a major component of global strategies aimed at providing appropriate health management protocols and biosecurity measures that protect national biological, social, and economic resources. At the same time, these support economically and environmentally sustainable aquaculture development, while also facilitating trade. Although the focus of this book and the chapter is to discuss pathogen related issues in aquaculture,

especially for cage aquaculture, a complete risk assessment is also necessary to guide acceptable levels of risk of a movement, covering assessment of ecological and genetic risks of such a movement. One recent example is the comprehensive transfer guidelines and risk management strategy developed by WorldFish facilitating safe transfer of GIFT from Malaysia to Nigeria (Amarasinghe, 2021; Arthur, 2021; Bartley, 2021; WorldFish, 2021).

Subasinghe *et al.* (subm.) discussed the non-conventional ways of dispersal of Harmful Aquatic Organisms and Pathogens (HAOP)¹ between ecosystems. HAOP may settle and reproduce beyond control to become pests in areas outside their original geographical distribution. The successful transfers are exacerbated by the changes in natural habitats (i.e., as a consequence of global warming, physical barrier/habitat destructions), facilitating and increasing species' direct transfer across natural boundaries. Unlike pollutions², HAOP often exhibit robust biological traits and can reproduce over time which makes their eradication almost impossible. This makes HAOP a challenging hazard to manage. Bio-invasions are seriously impacting aquatic ecosystems which are used by multiple industries, including aquaculture and fisheries. Therefore, science-based policies to protect marine ecosystems and the communities living on them must consider the specificity related to the risk of transfer and spread of HAOP.

11.5 Applying Biosecurity in Aquaculture

Applying practical and efficient biosecurity principles through well designed strategies in aquaculture requires an in depth understanding of the production system, including input supplies, service provisions, husbandry management practice, and the strength of the human capital. While good biosecurity in aquaculture production requires a holistic approach, especially in cage aquaculture, there are a few major aspects requiring strong emphasis.

11.5.1 Clean seed

Years of experience have now convinced the aquaculture industry and community that the use of clean and healthy seed should be given high priority in biosecurity for preventing disease outbreaks and subsequent losses (Subasinghe *et al.*, subm.). Clean seed requires a strategy for excluding pathogens. This specific pathogen free (SPF) strategy used in aquaculture began with Pacific whiteleg shrimp, *Penaeus vannamei*. Over the years, SPF shrimp jumped into industrial scale commercial operations taking the lead within the aquaculture industry and allowing the exponential growth of *P. vannamei* in Asia. The SPF strategy is now also applied in the salmon industry and is increasingly being embraced by other aquaculture stakeholders. Demonstrated benefits of applying SPF principles and technology in shrimp aquaculture for producing healthy seed have prompted research into “SPF Fish” by several companies and research groups. Invention of healthy “SPF Fish”, targeting for what we call “people’s fish” (i.e., carps, tilapias, and species of catfish, etc.), accessible and affordable to smallholders, will offer significant opportunities and benefits in the future for improving biosecurity in global aquaculture (Subasinghe *et al.*, subm.).

The use of genetic technologies in biosecurity goes beyond SPF seed. A recent breakthrough showed promise in combatting tilapia lake virus (TiLV) through selective breeding of fish with the resistance genes. In 2020, researchers from the University of Edinburgh's Roslin Institute and WorldFish analysed the genes of 1,821 Genetically Improved Farmed Tilapia (GIFT), which were tagged and placed in a pond that had an outbreak of TiLV. The variation in TiLV resistance were found to be independent of genetic variation in growth, meaning that any future breeding programmes for GIFT that produce fish resistant to TiLV will not adversely affect the growth of the fish, and will benefit farmers' yields. Use of TiLV resistant seed for tilapia in cage culture would be highly desirable and effective as producing fish in open waters pose serious biosecurity changes.

11.5.2 Clean feed

The biosecurity of aquatic diets lies in the quality (food safety) of the ingredients used and, in the way, ingredients and diets are stored. Mycotoxins as common contaminating infections of plant ingredients, produced from toxigenic fungi belonging to *Alternaria*, *Aspergillus*, *Fusarium* and *Penicillium*, among others, and those most encountered are as follows. Aflatoxin of which AFB1 produced by *Aspergillus flavus* is the most prevalent and toxic. A dose of 20 ppb in the diet presented to *Oncorhynchus mykiss* resulted in 56% of the stock developing hepatocellular carcinomas after 8 months and 83% after 12 months (Schoenhard *et al.*, 1981). In Nile tilapia, *O. niloticus*, doses of 100 ppb in the diet resulted in lower weight gain, FCR (food conversion ratio), hematological profiles, liver histopathology, elevated liver enzymes, and a lower resistance to *Aeromonas hydrophila* infection (Mahfouz and Sherif, 2015). Cyclopiazonic acid (CPA) a mycotoxin produced by *Aspergillus flavus* can depress weight gain in *Ictalurus punctatus* (Jantrarotai and Lovell, 1990). Deoxynivalenol (DON) produced by *Fusarium graminearum* at 0.3-2.6 ppm added to diets presented to *O. mykiss* for 8 weeks resulted in significant decreases in feed intake, weight gain, growth rate and FCR (Hooft *et al.*, 2011). Atlantic salmon, *S. salar*, given 0.5-6 ppm DON in the diet for up to 8 weeks inhibited protein synthesis, and growth. There is also a statistically significant dose-related reduction in antibody levels following vaccination against *Aeromonas salmonicida* (see Bernhoft *et al.*, 2018). Fumonisin produced by *Fusarium verticillioides*, and especially FB1 is among the most toxic of those produced. Doses of >80 ppm result in increased mortality (The Fish Site, 2013). Moniliformin (MON) from *Fusarium proliferatum*; diets containing >20 ppm in the diet results in reduced weight gain (Yildirim *et al.*, 2000). Ochratoxin A (OTA) is produced by species belonging to *Aspergillus* and *Penicillium*. *Ictalurus punctatus* fed 4 ppm ochratoxin A in the diet had kidney damage and lower weight gain (Manning *et al.*, 2003). T-2 toxin, a *Fusarium* mycotoxin, at low doses of 0.625 ppm depressed weight gain and lowered hematocrits while >1 ppm cause high rates of mortality (Manning *et al.*, 2005). *Ictalurus punctatus* fed either 4 ppm OTA or 1-2 ppm T-2 for 6 weeks and then subsequently challenged with *Edwardsiella ictaluri* on day-21 had statistically significantly ($p<0.05$) higher mortality than control groups (Manning *et al.*, 2005). The effects of zearalenone (ZEN), a *Fusarium* mycotoxin, accumulates in the ovaries with impacts reproduction in zebrafish, *Danio rerio* (see Schwartz *et al.*, 2010), the impacts on reproduction to other species like *O. mykiss* are unknown (Woźny *et al.*, 2013).

To address the biosecurity of aquatic diets, binders are added to prevent them from being absorbed. Hydrated sodium calcium aluminosilicate (HSCAS) clays work well in binding aflatoxins but less so for the other mycotoxins (Zhao *et al.*, 2010). Modified fractions of *Saccharomyces cerevisiae* are also reported to work well on a wider range of mycotoxins (Zhao *et al.*, 2010).

11.5.3 Vaccines

Vaccines are recognised as important tools for the prevention and control of diseases in fish. The number of fish vaccines commercially available have grown in recent years but there are still numerous key diseases where no vaccines are available or cases where existing vaccines do not perform well (see Table 11.1). Many vaccines are efficiently used in cage aquaculture, mostly among salmonids.

Commercial fish vaccines are available for a wide range of fish species (see Evensen, 2016), including Atlantic salmon, rainbow trout (*O. mykiss*), sea bass (*D. labrax*) and sea bream (*S. aurata*), tilapia (*O. niloticus* / *O. mossambicus*), amberjack (*Seriola dumeril*) and yellowtail (*Seriola quinqueradiata*) in Japan, catfish (*I. punctatus*) and Vietnamese catfish (*Pangasionodon hypophthalmus*). Many scientists and vaccinologists have made significant progress towards producing a commercial vaccine against tilapia lake virus (TiLV) and an affordable and accessible vaccine will shortly be available.

Effective improvements on current vaccines are still required for some of the traditional inactivated vaccines on efficacy (identification and optimisation of antigen components), choice of adjuvant and oral administration. Development of successful vaccines against intracellular bacterial pathogens and viruses may require the use of live attenuated vaccines and application as oral vaccines, although there are safety concerns with live vaccines. Vaccines against parasites and fungi are also in development and these may need to rely on recombinant or DNA vaccine technology. There is also still a requirement for basic information on pathogenesis, immune response, and identification of potentially protective antigens for such parasites and fungi (Adams and Subasinghe, 2021).

11.5.4 Diagnostics

As finfish aquaculture continues to expand there is a commensurate need for improvements to aquatic biosecurity practices and in the management of aquatic animal health and in the support that can be offered at the individual farm level. This can be done partly by improving the capabilities of individual farmers to carry out health checks in “real time” (i.e., within a time frame that allows effective decision-making on therapeutic or preventative actions). Point of care diagnostics (POCDs) are tests that are designed to be used on site to provide rapid results without the need for dedicated laboratory facilities. Examples of POCDs include water quality test kits, pH strips, bacteriology testing, lateral flow immunoassay, loop-mediated isothermal amplification (LAMP), sentinel water quality probes – for the measurement of potentially deleterious nitrogenous wastes, oxygen, and temperature probes, etc. Even basic POCDs can facilitate decision-making on the health status of animals without the delays associated with conventional laboratory

testing. POCDs can be used in investigation of disease outbreaks; in passive and active health surveillance for pathogen screening; as an early warning system to prevent disease outbreaks; and, in certification for animal movements.

To ensure the accuracy and consistency of results, individual POCDs need to be validated and approved by the relevant national competent authorities. As each POCD may have different sensitivity and specificity for specific diseases, their application and interpretation of results should also be considered.

POCDs should be portable, easy to operate ("user-friendly"), and the cost of each test should be low. Obviously, devices should be sufficiently accurate, specific, and sensitive to facilitate early-stage detection of an infection and improved decision-making to avoid costs associated with advanced infections. POCDs can play an important role in remote settings and in resource-limited farms where access to disease diagnostic services is limited and where practical considerations regarding what diagnostics can be conducted "pond-side" or in basic farm laboratories. POCDs can also facilitate the correct application of treatment regimens (drug, dose, and time), thereby reducing the indiscriminate (and ineffective) use of drugs and chemical.

Results of POCD's are interpreted in context when making a diagnosis. Test results are one element in making a disease diagnosis and should not be considered as definitive or taken in isolation. The application of POCDs in human medicine is often at the level of the individual patient whereas in aquaculture they are usually applied at a population level). Thus, the final diagnosis should be made by a health professional based on the POCD results and clinical signs, if available.

While POCDs can be effectively applied at farm level to improve diagnostic capability they should be considered within the context of health management at farm level. It is important to realise the critical roles played by veterinarians and health diagnosticians in developing and managing local, regional, and national health care strategies and the role of POCDs within this framework.

POCDs should also be viewed within the complete framework of diagnostic methods used in health management. In addition to POCDs, it is important to consider physical and behavioural assessment of individuals and populations, microscopic examination of tissues and body fluids and environmental parameters (e.g., water quality parameters including harmful phytoplankton / zooplankton) when making a presumptive diagnosis.

As aquaculture continues to intensify and develop, there will also be an increased risk of infection from existing and exotic pathogens. While many advanced aquaculture production industries move towards greater degrees of precision farming with the employment of a suite of tools that integrate artificial intelligence (AI), image analysis (IA) and big data-based technologies to support biosecurity and aquatic health management, these developments are not, currently, a global requirement across finfish aquaculture. For most producers, represented by small to medium-sized farms, there is a greater need in having increased accessibility to POCDs and veterinary support empowering farmers to make real-time management decisions. In support of this, national governments might consider

setting standards for aquaculture diagnostics including what testing should be conducted, the approval of standard methods/POCD devices, stating who is authorised to do the testing, and setting standards for training of animal health workers. Commercial development of solutions should be encouraged within a health management framework to reduce the burden on government services, drive innovation and reduce costs for farmers. There needs to be a parallel development of the reporting system to capture and process disease diagnostic data and communicate it back to the farming community to provide timely warnings of potential disease threats and enable national health agencies to target their activities (e.g., in disease containment or in the deployment of support services, etc.) more effectively. This reporting system must incorporate disease diagnostic data from public and private sources such as aquaculture veterinarians and trained health diagnosticians. It should also include a concerted communications strategy from national health authorities to farmers stressing the importance of approved diagnostic tools for the surveillance / testing of the major pathogens of concern.

For POCD devices, improvements in operational sensitivity, accuracy, analysis run times and unit cost of pathogen testing will ultimately determine their rate of uptake, use and benefit to farmers. There will be a need for multiplexing – multiple pathogen screening capability and for the adoption of sensitive eDNA-like based technologies to assess the health of the entire population or of systems rather than a small number of specimens sub-sampled from it. It is expected that the next decade will see further developments in virtual veterinary diagnostics supported by smart phone with mobile healthcare packages or software applications to: capture data, images, video that feed into centralised health support labs to allow the health, welfare and behaviour of stocks to be assessed; provide early-warning of disease threats relevant to discrete water catchments/regions; use data that feed into trained neural network / artificial intelligence-based decision making systems; permit track and trace of diseases; and, in the provision of on-line health training. As with POCDs themselves, it is important that such approaches are validated and accredited.

11.6 Biosecurity Governance

Biosecurity governance is complex but necessary to ensure efficient implementation of biosecurity plans and strategies. The global nature of aquaculture has created national, regional and/or international frameworks comprising treaties, agreements, declarations, guidelines, and policies. However, the binding (obligatory) and non-binding (voluntary) nature of these instruments have reduced the efficacy of biosecurity implementation and management worldwide. Interestingly, most international, and regional biosecurity frameworks (57%) do not require compliance, and all others (43%) require compliance only when ratified by a nation state (Campbell *et al.*, 2020).

National legislation and regulatory frameworks are critically important for sound biosecurity. There should incentives for good compliance while disincentives for bad compliance. National legislation must project the interest of all actors along the aquaculture value chain. However, legislation is of no value if it is not adequately enforced.

Administration of shared water bodies under different national jurisdiction, which may challenge biosecurity and disease control, requires strong coordination and collaboration among states, countries, and authorities. Harmonised legislation and enforcement procedures are equally important. In this regard, various frameworks covering groups of countries based on political and/or geographical coverages exist (APEC/FAO/NACA/SEMARNAP, 2001; OIE, 2021a). However, the effectiveness of such arrangements in protecting geographies from incursion and spread of pathogens remain questionable.

11.7 Progressive Management Pathway for Improving Aquaculture Biosecurity

While many efforts have been exerted by national competent authorities, industry, and the academe, as well as regional and international entities, and development institutions, successful biosecurity practices have not been properly implemented in many areas, so actions have been reactive or *ad hoc*, which is significantly more costly than investment in preventative measures (Subasinghe *et al.*, *subm.*).

The Progressive Management Pathway for Improving Aquaculture Biosecurity (PMP/AB) was developed by the Food and Agriculture Organization of the United Nations (FAO) and partners, as a 'paradigm shift' after analysing the pathways and factors to disease emergence and seeing a need for strategic planning to further guide and support countries towards achieving sustainable aquaculture biosecurity and health management systems. The PMP/AB is an extension of FAO's "Progressive Control Pathways" (PCP) approach, which has been internationally adopted to assist countries develop risk mitigation strategies that reduce or prevent losses from major livestock diseases. However, whereas most PCPs focus on control of specific diseases, the PMP/AB focuses on diseases faced by aquaculture at the commodity and enterprise level. The PMP/AB uses a comprehensive, holistic approach to improving aquaculture biosecurity and supporting sustainable development.

The PMP/AB is progressive, collaborative, and risk based (see Figure 11.6). The four stages of PMP/AB involve strong public and private stakeholder input to promote the application of risk management at the sector level as part of a national approach. Countries decide the appropriate entry-point, how far and how fast to progress to the next stage. Due to the wide variation of species farmed in each country, aquaculture sectors may advance independently, at different speeds or with different goals but a common requisite is strong cooperation between government and industry, such as a public-private sector partnership (PPP). This is necessary to ensure clarity on roles and responsibilities, identify key gaps requiring improved capacity and infrastructure, and increase awareness of the cost/benefits of biosecurity along the whole value chain. Risk analysis is a key aspect of all stages of the PMP/AB. Risk hotspots (critical control points) are identified for biosecurity investment (training, diagnostic capacity, etc.). All this feeds into development of a National Strategy on Aquatic Animal Health (NSAAH) or national aquaculture biosecurity strategy, which sets the foundation for ongoing review and updating as the industry develops.

[Figure 11.6 – about here]

The benefits of the PMP/AB can be summarised as follows:

- It builds on management capacity using bottom-up and top-down approaches, is evidence-based and supported by transparent and ongoing review. The co-management approach ensures problems are clearly defined and management solutions have common understanding and buy-in.
- The PMP/AB provides a degree of consistency between participating countries or regions that is essential for reducing risks from trade, as well as addressing biosecurity-related trade challenges.
- The PMP/AB provides a framework that is adaptable and can respond to changes in aquaculture production scope and objectives (small to large-scale; local to international industries), as well as to environmental and anthropological changes that impact aquaculture production.

11.8 Climate Change Impacts on Diseases in Aquaculture

Climate change effects on aquaculture production and their sustainability implications, mitigation, and adaptations have been recently reviewed by Maulu *et al.* (2021) and a general account on the impacts on cage aquaculture is provided in Chapter 1 (this volume). Most of the climate changes occurring in the world, both in the north and in the south, appearing to impact aquatic environment and its life, affecting aquatic production and aquaculture. However, hard evidence on the scale of climate change impacts on aquaculture and their socio-economic and environmental implications are yet to be understood.

Fish health is governed by the quality of its culture conditions, including temperature, oxygen solubility, salinity, the content of dissolved nitrogenous waste products, etc (Mydlarz *et al.*, 2006) and adverse environmental conditions or rapid changes can negatively impact welfare. Complex inter-relationships exist between many chemical and physical parameters in the aquatic environment where changes in one parameter can affect the toxicity of others, with consequential impacts to fish health.

Marine cage sites are frequently positioned in “shallow” coastal waters and estuaries and are susceptible to changes in water temperature, storm intensity or frequency and sea level. The biosecurity of exposed sites can be threatened by wave action and are susceptible to storm damage. Wave energy is greatest at the surface and as waves approach the coast, the velocity of the waves decreases but their height increases. The severity and frequency of storms have been predicted to increase (Frost *et al.*, 2012), resulting in potential structural damage to fish enclosures with consequential breaches to biosecurity in either the loss of stock or through the introduction of disease agents, vectors, and predators with the movement of flood waters.

For pen sites situated in rivers and estuaries, there concerns regarding the frequency of flood events, their magnitude, the changes to hydrology and the consequential biosecurity impacts associated with run-off and the increased transport of pathogens, sediments, pollutants, and nutrients (Callaway *et al.*, 2012).

According to Collins *et al.* (2020), diseases in aquaculture are likely to be affected by a changing temperature regime, but in a largely unpredictable manner. However, what is certain is that when fish are exposed to thermal stress, they become more susceptible to diseases and that warmer conditions may result in the establishment of exotic diseases. Since fish become more vulnerable to diseases and pathogens through both direct and indirect climate impacts, especially through thermal stress (Chiaramonte *et al.*, 2016), and thus, warm water disease outbreaks are predicted to occur more frequently in addition to the possibility of discovering new ones under a changing climate (Sae-Lim *et al.*, 2017).

Logically, at least up to certain limit, rising temperature will accelerate the replication rate, virulence, life-cycle longevity, and transmission of pathogens in fish, promoting emergence of diseases production systems. In salmon cage culture, warm water pathogens, such as sea lice will remain a challenge and further warming is likely to worsen the infections in cold temperate conditions, requiring more treatments thus more expenses (Collins *et al.*, 2020). Undoubtedly, increased disease incidents in aquaculture production systems will lead to reduced profits, consequently, affecting the socio-economic viability of aquaculture. In the contrary, rising temperature may positively affect cold water diseases, such as vibriosis and winter ulcer that affects Atlantic salmon and might gradually become extinct due to the emerging unfavourable conditions (Sae-Lim *et al.*, 2017) which may favour aquaculture production.

After water, oxygen is the first limiting component of the aquatic environment. Oxygen solubility decreases with increasing salinity and temperature; the solubility of oxygen in seawater is approximately 20% lower than in freshwater. Pörtner and Knust (2007) warns that the thermal limitation of oxygen will be among the main physiological effects of climate change. This has been examined by Daufresne *et al.* (2009), Baudron *et al.* (2014) and by van Rijn *et al.* (2017), among others, who highlight the 5-20% decline in fish body size over recent decades in response to changes in global temperatures. For some aquaculture fish species, rises in temperature may push them beyond their optimal or tolerant culture ranges (see Table 1 of Jahangiri *et al.*, 2021), may have impacts on egg equality and production (King and Pankhurst, 2004), their capacity to tolerate environmental stresses and/or their susceptibility to infections. The immune function of some farmed fish species may increase within their optimal temperature range (Bly *et al.*, 1997; Bowden *et al.*, 2007), but this may be compromised by the increased stress associated with lower dissolved oxygen concentrations in warmer waters. Increased water temperatures may also result in the increased loading of pathogens within sites and extend periods of exposure, potential susceptibility, and infection (Harvell *et al.*, 2002).

It is well known that harmful algal blooms occur due to change in climatic conditions. They are a serious threat to the environmental sustainability of aquaculture production. For example, some flagellates and dinoflagellates are potentially toxic or nuisance species which could be responsible for stress or mortality among finfish and shellfish (Gubbins *et al.*, 2013). Trainer *et al.* (2019) reported an unprecedented loss of fish ever recorded worldwide in Chilean aquaculture due to the expansion of *Pseudochattonella* and *Alexandrium* species

whose outbreak were associated with climate-induced changes in water column stratification. Some studies have also confirmed pathologies, such as inflammation, atrophy, and necrosis in several organs of bivalve molluscs resulting from harmful algal blooms (Haberkorn *et al.*, 2010; Basti *et al.*, 2011; Hégaret *et al.*, 2012). Although there is limited information on the mechanisms through which climate change will affect toxic substances in aquaculture, Farrell *et al.* (2015) reported that temperature variation can affect the metabolism of most widespread harmful algae.

While bad weather events and damage to culture infrastructures account for significant percentages of non-pathogen associated stock losses, predators also account for a high proportion of losses. Predators include birds (EIFAC, 1988; Kumar *et al.*, 2021; Radwan, 2022), fish (Bevan *et al.*, 2002), sea lions and seals (Tillapaugh *et al.*, 1993; Nash *et al.*, 2000; Sepúlveda and Oliva, 2005; Vilata *et al.*, 2009; The Fish Site, 2021), otters and mink (Bruun-Schmidt *et al.*, 2000; Bevan *et al.*, 2002), sharks (Angel and Edelist, 2013; Ceyhan *et al.*, 2020), etc. Predators can damage pen enclosures and can impact on production directly resulting in mortalities; can inflict damage of sufficient severity that they are subsequently lost from production; can result in non-lethal wounding, transferring pathogens or creating wounds that are subject to secondary infection; or can result in stress reducing production performance (Bevan *et al.*, 2002). Predators can be excluded by culturing stock within enclosed systems (e.g., greenhouses or enclosed netted enclosures using a small size mesh to keep out aerial predators), anti-predator nets around submerged pen enclosures or by using overhead wires to keep out birds. For larger sites or where the former is not possible, then predators can be deterred by changing the design of the culture site (e.g., using ponds with an increased water depth, steep banks, removing bird perching structures) or by using lights, scarecrows, sonic propane cannons, alarms, water spray devices, reflectors, silhouettes of predatory birds, or by having dogs and personnel on site (Bevan *et al.*, 2002). Constant sound emitting acoustic deterrent devices (ADDs) have now been discontinued due to the potential disturbance that they can cause to other marine mammals. Instead, the next generation of artificial intelligent, night, and thermal-imaging camera devices - Response (ASR) and Electric Startle Response (ESR) devices which recognise approaching predators and then emits a randomised frequency of pulses to deter them (Fish Farming Expert, 2021). The use of fibreglass sound-emitting orcas on the Isle of Lewis, Western Isles, Scotland, have been used to scare away seals where historical seals have accounted for a 1% mortality of fish (Fish Farming Expert, 2019).

11.9 Future strategies for improving aquaculture biosecurity

Sievers *et al.* (2022) reviewed the biological challenges and opportunities of submerged cage aquaculture of marine fish. They concluded that submerging aquaculture cages hold the promise of providing relief from periods of less-than-optimal environmental conditions, reducing fish interactions with harmful organisms, and unlocking new production areas devoid of conflict with other coastal users. They also noted that not all fish species will be similarly suited to submerged culture, and a suite of key challenges and bottlenecks stand in the way of commercial production of several species. If submerged culture is to mature and fulfil its promise, research to empirically document production and environmental

benefits, and issues surrounding fish welfare throughout the production cycle needs to lead the way.

Earlier in this chapter, we mentioned the importance of POCDs. While POCDs could help smallholders with limited budgets, for industry stakeholders with larger biosecurity budgets, a suite of advanced technologies is employed across the production pipeline to monitor and safeguard the health of stock and culture systems. These include, as examples, systems for: 1) fish recognition / detection systems (e.g., You Only Look Once (YOLO) real time object detection models; Xu and Matzner, 2018; Stavelin *et al.*, 2021); 2) in the calculation of stock numbers, biomass, growth performance, and for grading (Lopes *et al.*, 2017; Zhao *et al.*, 2021); 3) sensors estimating appetite, detecting hunger levels and ensuring diet delivery in all weathers (e.g., Anderson, 2015; Måløy *et al.*, 2019; The Fish Site, 2022) and in reducing feed waste (e.g., Observe Technologies, 2022); 4) behavioural analysis of pen-reared stocks using recurrent neural networks (e.g., Zhao *et al.*, 2018; Romero-Ferrero *et al.*, 2019); 5) health status assessments, e.g., health and survival (Chang *et al.*, 2021), of salmon skin health (Sveen *et al.*, 2021), and for screening for sea-lice (e.g., counting from Moore, 2020) and their removal (e.g., Stingray Marine Solutions AS, 2020); 6) predicting disease outbreaks (e.g., of sea lice using AquaCloud, see NCE Seafood Innovation, 2017), including phytoplankton and algal blooms (Zhang *et al.*, 2016; Holmyard, 2019); 7) monitoring water quality through the use of, e.g., AI, satellite remote sensing technologies, sensor equipped drones and sentinel probes to collect water quality data (Ubina and Cheng, 2022); 8) improving farm efficiency, e.g., aerators, pump flow, etc. (Lee, 2000; Manoharan *et al.*, 2020); 9) monitoring environmental impacts (e.g., in assessing bacterial and ciliate populations around cages) through the analysis of eDNA barcodes (e.g., Frühe *et al.*, 2020); 10) predator recognition (Fish Farming Expert, 2021); and 11) fish breeding programmes using AI algorithms to increase the prediction accuracy for disease resistance (tolerance or resistance) (Bargelloni *et al.*, 2021; Vu *et al.*, 2022).

In diagnostics, advanced tools include the development of lab-on-a-chip / microarray diagnostics for a suite of aquatic pathogens, including Min-ION-like technologies (Gallagher *et al.*, 2018; Delamare-Deboutteville *et al.*, 2021), eDNA approaches (Bastos Gomes *et al.*, 2017), microfluidic based technology using layered paper to control fluid flow (see Martinez *et al.*, 2010) etc. MinION, for example, allows for the multiplex, rapid high throughput sequencing of samples but its utility requires a parallel development in the rapid analysis of bioinformatic data.

Precision tools alone are not sufficient to make an accurate diagnosis. Trained human resources are equally important to ensure accurate diagnosis. Access to qualified aquatic veterinary health care provision in many countries is a challenge; the number of providers may be limited, or expertise covering the full breadth of species and health issues encountered within the country may not exist. In such cases, the use of POCDs can provide significant benefits in health care of aquaculture stocks. However, there are also significant challenges, not least in demonstrating to farmers the economic benefits of investing in POCD's and in training of farmers, technicians and local health practitioners on their proper use and application.

11.10 New and Next-Generation Farm Systems

In addition to the implementation of intelligent technologies within culture systems to directly address disease (i.e., systems for the early detection of disease conditions or to remove pathogens in waters where they are currently found), many of the challenges from disease in traditional flow-through systems have led to the design of systems to exclude pathogens. These include the redesign of existing sites to closed water systems, managing water quality on site, thereby negating the need to abstract water from common water pool systems where the disease threats are potentially higher and unknown. Such systems may include the use of Recirculatory Aquaculture Systems (RAS) or the move to land-based farming systems (e.g., there were >75 landed salmon farms in 2020; Cox, 2020). At the same time, some enterprises have looked to offshore farming to avoid some of the challenges associated with coastal aquaculture (lower water quality; higher prevalence of certain pathogens, e.g., sea lice; availability of new site permits, etc). Among the most notable are SalMar's 250,000 m³ volume Ocean Farm 1 (SalMar, 2021; see Figure 11.7) and Nordlaks' 385-metre long Havfarm 1 which has capacity to produce 10,000 tonnes of salmon (Soltveit, 2020). The design of several closed-containment production units also looks to address problems of sea lice infection, predator attacks and escapees. The use of net snorkels and skirts ("anti-sea lice shields") made of a fluid permeable polyester material are used around salmon cages to reduce sea lice infections by 80% (Dickie, 2018; Stein *et al.*, 2018). Nekkar's "Starfish" is a closed cage system (Nekkar, 2021), Hauge's 1850 m³ capacity "Egget" (Hauge Aqua Solutions AS, 2016) and FishGLOBE's 3,500 m³ enclosed salmon culture units protect stock from sea lice. Plans to increase the size of the FishGLOBE units to hold 30,000 m³ of salmon are currently being explored through industrial partnership (Salmon Business, 2022).

[Figure 11.7 – about here]

11.11 Closing Remarks

If aquaculture is to help meet the demands for animal proteins of a growing population in the face of climate change, then substantial gains in production can be made by first, reducing the magnitude of current loss attributable to disease due to lapses in biosecurity and secondly, by adapting culture systems which meet the necessary biological demands for each species. This chapter highlights the importance of health surveillance in the early detection and treatment of disease. While high levels of site biosecurity may require high levels of investment, if not appropriately maintained, when there are breaches of biosecurity in these sites, the magnitude of stock and economic losses can be large (i.e., due to the size of culture units, stocking densities used, etc). Some aquaculture stakeholders are moving towards greater degrees of precision farming using, for example, a suite of AI tools based as part of their biosecurity protocols for monitoring stocks, their behaviours and health status. For the remainder, the most substantial part of global aquaculture production, is represented by farming conducted in lower investment systems. As such the probability of disease events in these systems is higher but the magnitude of economic loss is typically lower. While the onus of implementing robust biosecurity practices is frequently heaped upon the grow-out farmers,

biosecurity is a responsibility throughout the entire production chain, which needs to operate within regulatory frameworks, enforced by law, to support sustainable production. FAO recently stated that “a paradigm shift is needed in dealing with aquaculture biosecurity risks”. A subsequent review of biosecurity across the specific segments of the aquaculture value chain conducted by Subasinghe *et al.* (subm.) calls attention to the need for efficient and effective biosecurity strategies and protocols at all levels in the next decade. Emphases should be on the need for: (a) emergency preparedness and response to disease outbreaks, (b) diagnostics, (c) microbial management at production level, (d) disease and pathogen surveillance, (e) welfare, (f) regulations and legislation, (g) fish trade, (h) research and technology development, (i) antimicrobial resistance, (j) the identification of non-conventional means of pathogen transfer, and (k) in training and access to health support networks.

Notes

¹ Wording established by Article 1 of the International Convention for the Control and Management of Ships’ Ballast Water and Sediments, 2004 (also known as the Ballast Water Management Convention or BWM Convention).

² The United Nations Convention on the Law of the Sea distinguishes ‘pollution of marine environment’ (i.e. ‘[...] the introduction by man, directly or indirectly, of substances or energy into the marine environment [...]’; Article 1, #4) from biohazards (Article 196).

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Figure 11.1. The Nile tilapia, *Oreochromis niloticus*, cultured at this undisclosed Nile tilapia riverine site were reared in cages 12 units deep and in blocks which ran for several hundred metres without a break. The poor water exchange, water quality, stress and bacterial problems that were created led to a significant loss of stock. Image A. Shinn.

Figure 11.2. Pathogens of cage-reared finfish.

a. *Myxobolus* sp. from Vietnamese striped catfish, *Pangasianodon hypophthalmus* (image by A. Shinn); b. Adult female *Ergasilus sieboldi* on the gills of a rainbow trout, *Oncorhynchus mykiss* (image by A. Shinn); c. Adult female *Lepeophtheirus salmonis* on Atlantic salmon, *Salmo salar* (courtesy of D. Conway); d. Feeding activity of *Caligus epidemicus* (*) has denuded the epithelium of a brown-marbled grouper, *Epinephelus fuscoguttatus* (image by A. Shinn); e. Gravid female *Argulus foliaceus* (image by A. Shinn); f. Strobila of *Eubothrium crassum* (courtesy of D. Conway); g. Metacercariae of *Diplostomum spathaceum* released from the lens of rainbow trout (image by A. Shinn).

Figure 11.3. Diseases and pathogens of cage-reared finfish.

a. Dual exophthalmia in a Nile tilapia, *Oreochromis niloticus*, infected with *Streptococcus agalactiae* (image by A. Shinn); b. An *Aeromonas hydrophila* infection in Nile tilapia; c. Brown trout, *Salmo trutta*, infected with *Saprolegnia* sp.; d. *Apiosoma* sp. and *Gyrodactylus* sp. (*) infection (courtesy of G. Paladini); e. *Ichthyobodo* [*Costia*] *necatrix* on the skin of a brown trout, *Salmo trutta* (courtesy of G. Paladini).

Figure 11.4. Building biosecurity into the infrastructure of production sites. The design of a site should minimise cross contamination between different sections; permit the management/ treatment of water flow through the site; and, permit each production unit to be effectively disinfected on a routine basis. Image courtesy of Fish Vet Group Asia, Benchmark.

Figure 11.5. Main risk pathways for pests and diseases to spread on to, within and from a farm. Inserts (by A. Shinn), adapted from NZMPI, 2016.

Figure 11.6. Proposed FAO implementation of aquaculture biosecurity, using the Progressive Management Pathway (PMP-AB). Adapted from FAO (2019).

Figure 11.7. Owned and operated by SalMar, Ocean Farm 1 is the first salmon farm designed and built for exposed operation, with a novel design that combines solutions from aquaculture and the offshore industry. (<https://www.dnv.com/news/ocean-farm-1-receives-first-ever-offshore-fish-farming-class-certificate-from-dnv-gl-165626>).



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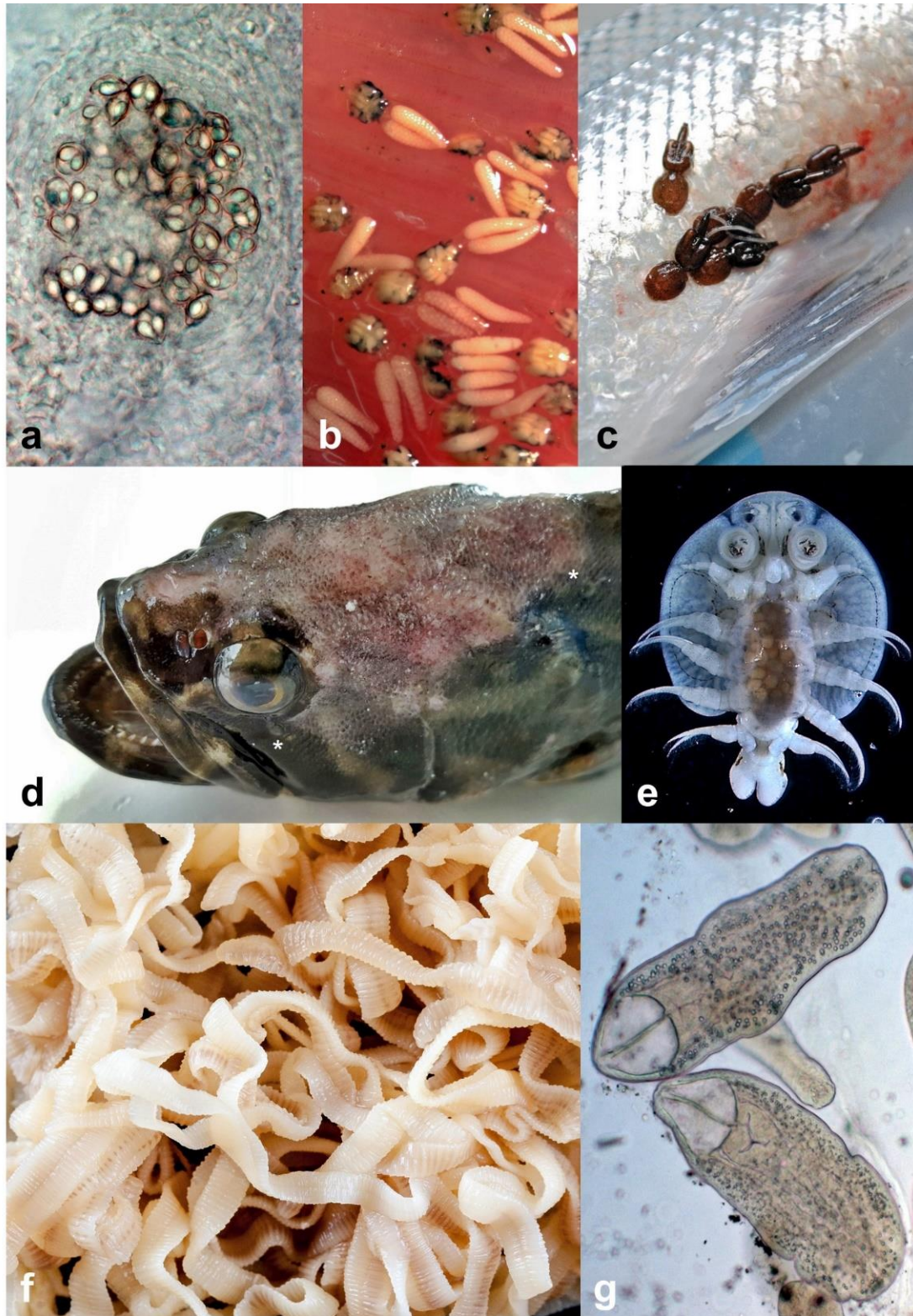


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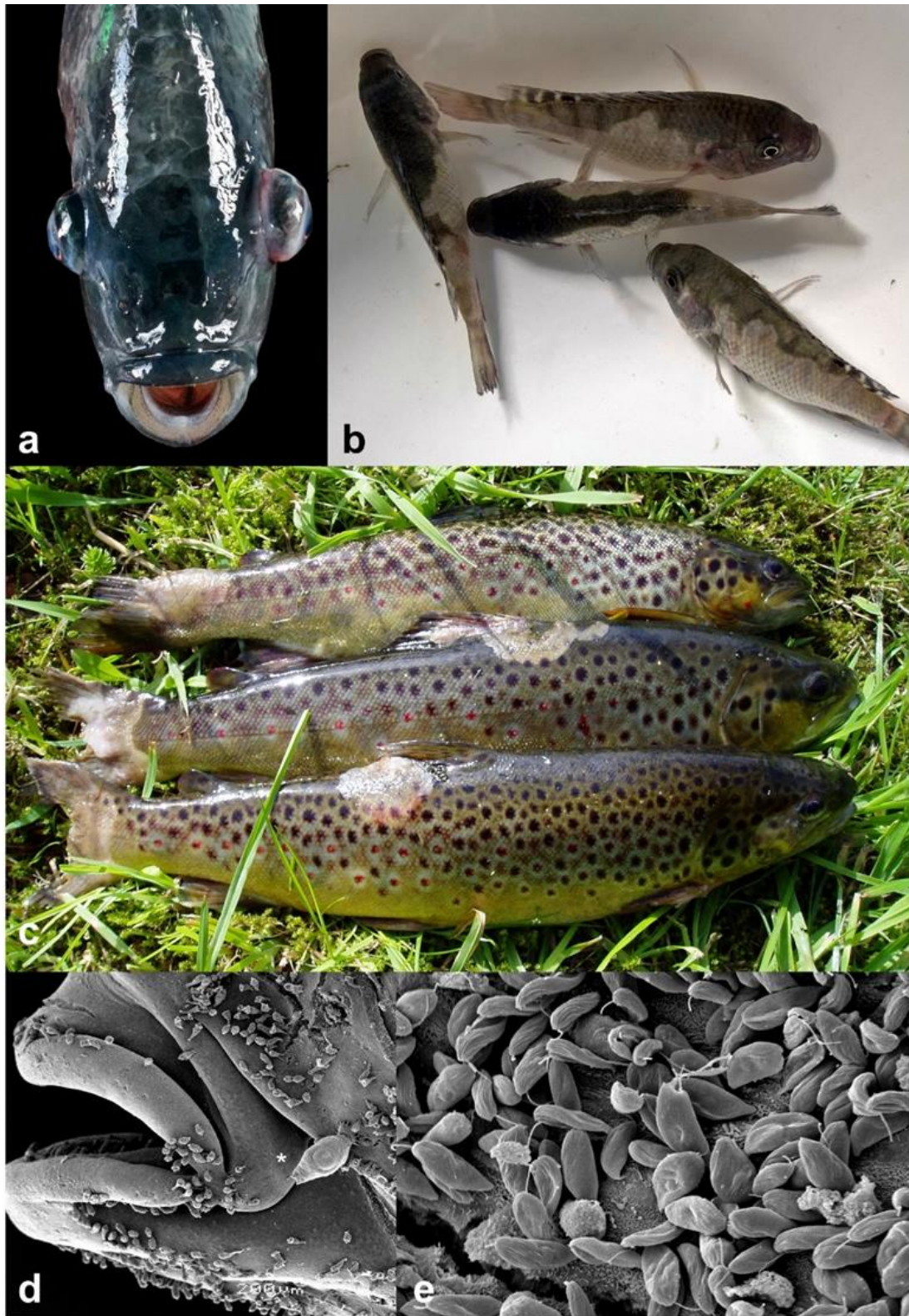


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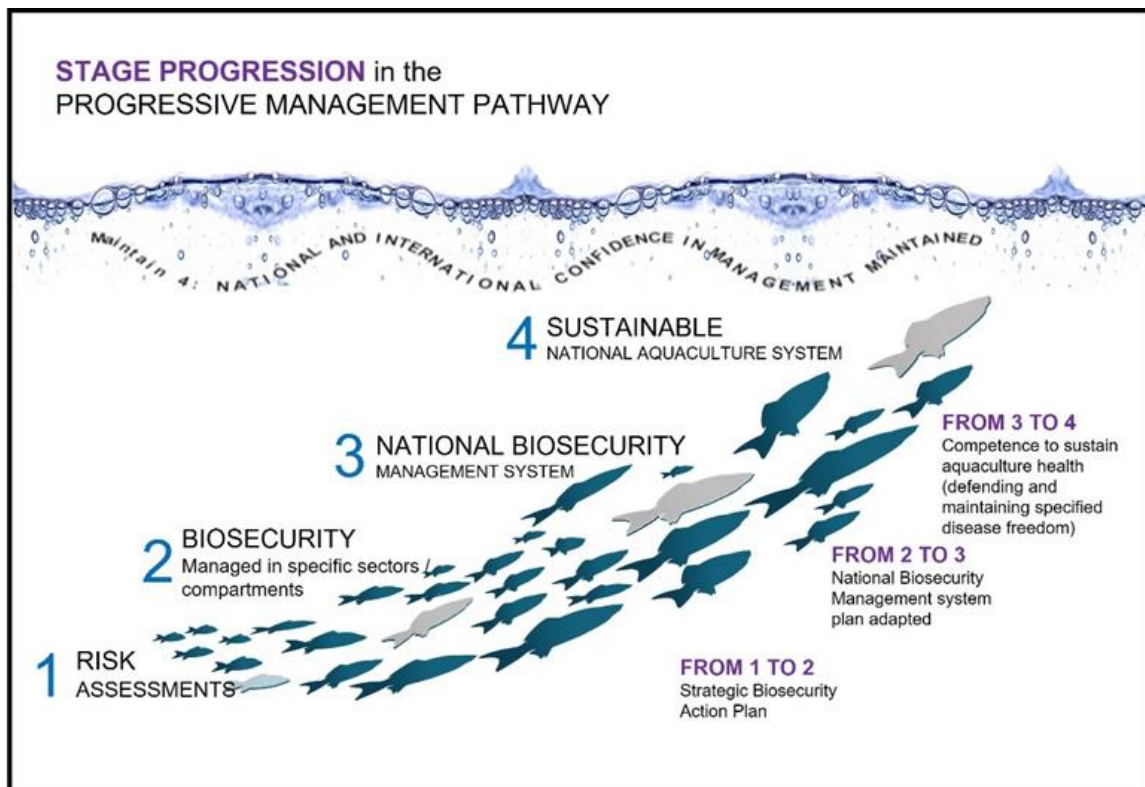


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